

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Potential Avenue of Attack



Learning Outcome:

Understand information about relational database management system in spoken or written texts.

Ask your partner:

- What do you know about vulnerability in computing?
- Have you ever found a security risk in computing?

Can Do Statements:

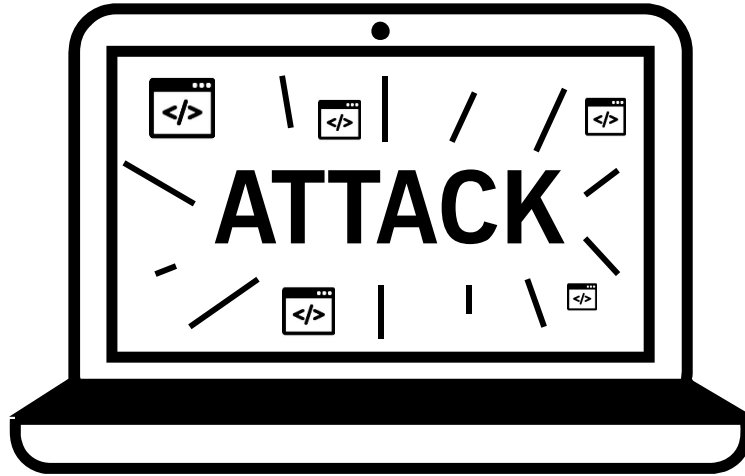
- I can read information about the concept of vulnerability in computing.
- I can use terms and expressions related to the concept of vulnerability.
- I can write a text to summarize the risks of vulnerability in an organization.





Let's Explore!

You have to evaluate potential vulnerabilities frequently.



Instructions: Read the text below.

In computer science, there is an issue called vulnerability that affects the software security industry. A product is considered vulnerable when it has a security exposure, this means a weakness. This weakness is not supposed to be introduced by the product developer, but it is important to identify it and fix it as soon as possible in order to decrease the system information assurance. In fact, the one who spots a product weakness is called an attacker, a person who gets into the system by doing so. However, the concept of vulnerability should not be confused with anything that causes an attack against the system because in some cases it refers to bad user's usage of the product.

Whenever a weakness is detected in a product, it compromises aspects like the availability, integrity, or confidentiality of a product. Availability refers to the capacity to access a product or resource; in other words, the attacker is not supposed to control and cause the server to fail. The system should be able to detect and block this action on time. On the other hand, integrity refers to trustfulness; the attacker is not supposed to modify any of the elements of the product without authorization. Plus, another goal for security is confidentiality, which refers to limiting the access of the information that must be kept in private. All in all, the attacker is not supposed to compromise any of the data within the product.

If an organization is established, it should know that vulnerability has two possibilities: with risks and without risks. Risk can be understood as a chance of loss. When there is vulnerability with risk, the company ought to protect the data and avoid the security defects. However, when there is vulnerability without risk, it means that the defect was already fixed or removed. Anyway, the company ought to find the way to disable the attacker as fast as possible.

**Let's Practice!**

The concept of vulnerability.

Instructions: Read the paragraph. Underline all the mistakes.

In computer science, we are not opposed to take anything for granted, specially taking into account the vulnerability that affects the software security industry. A product bust be considered vulnerable when it has a security exposure. This weakness his not supposed to be introduced for the product developer; that is why we aught to identify it and fix it as soon as possible in order to decrease the system information assurance. The concept of vulnerability shoulder not be confused with anything that causes an attack against the system because, in some cases, it refers to bad use of the product.

**Let's Practice!**

Important concepts related to vulnerability..

Instructions: Use the words from the box to correctly complete the sentences.

Word Bank

should / to be / is not supposed / are not supposed /
are supposed

1. The one who spots a product weakness is supposed _____ called an "attacker".
2. "Attackers" _____ to get into the system.
3. These weaknesses _____ to compromise any important aspect like availability, integrity or confidentiality of a product.
4. Remember that the attacker _____ to control and cause the server to fail; it should be a real risk for security vulnerability.
5. Integrity refers to the trustfulness; the attacker _____ not modify any of the elements of the products without authorization.



Extension Activity

Talking about computing vulnerability.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Make a presentation about the risks of software security vulnerability for any organization.
- Create some slides or poster to support your presentation.
- Include terms and expression in relation with the concept of vulnerability.
- Show your presentation to the class, be ready to ask and answer some questions

Self-Assessment

Yes No

- I define the concept of vulnerability in computing.
- I recognize different terms related to vulnerability.
- I make a presentation summarizing the risks of vulnerability in an organization.

☐	☐
☐	☐
☐	☐



Target Words

- **vulnerable** (adjective): something that has a weakness and is easy to assault due to lack of defense
e.g. A product is considered vulnerable when it has a security exposure, this means a weakness.
- **assurance** (noun): something that can be guaranteed or provides security
e.g. This weakness is not supposed to be introduced for the product developer but it is important to identify it and fix it as soon as possible, in order to decrease the system information assurance.
- **disable** (verb): to make someone stop what he or she is doing
e.g. The company ought to find the way to disable the attacker as fast as possible.
- **authorization** (noun): to give permission or authority
e.g. Integrity refers to the trustfulness; the attacker is not supposed to modify any of the elements of the products without authorization.
- **loss** (noun): a chance to fail what you possess
e.g. Risk can be understood as a chance of loss.
- **trustfulness** (noun): something that can be trusted
e.g. Integrity refers to the trustfulness; the attacker is not supposed to modify any of the elements of the products without authorization.



Learning Points

Be Supposed to in Negative Statements

You use **be supposed to** when you have to do something or it is an obligation.

In affirmative statements, you use **be supposed to** to express something that it is intended to happen or similar in meaning to "It is said"

Example:

*Companies **are supposed to** know how to protect themselves from attackers who can access their information.*

In negative statements, you use **be not supposed to** for something that is not allowed or basically prohibited. You use the verb to be form in singular or plural when corresponding.

Example:

*The product **is not supposed to have** flaws and defects.*

*The attackers **are not supposed to control** the software.*

Examples

- *This weakness is not supposed to be introduced for the product developer but it is important to identify it and fix it as soon as possible in order to decrease the system information assurance.*
- *The attacker is not supposed to control and cause the server to fail, it should be a risk for security vulnerability.*
- *The attacker is not supposed to modify any of the elements of the products without authorization.*



Learning Points

Modal Verbs for What is Advisable

You use **modal verbs** like **ought to** and **should** to give advice and make suggestions.

An advice might be also known as an opinion or recommendation. The idea is to conduct or guide.

In software some suggestions must be followed in order to avoid fraud, flaws or defects in a product.

In affirmative statements, the use of should and ought to give advice in a polite way.

Example:

In the company, you should identify the security holes in order to protect your company and your employees.

- You ought to protect your personal information.

Examples

- *The concept of vulnerability should not be confused with everything that causes an attack against the system.*
- *When there is vulnerability with risk, the company ought to protect the data and avoid the security defects.*